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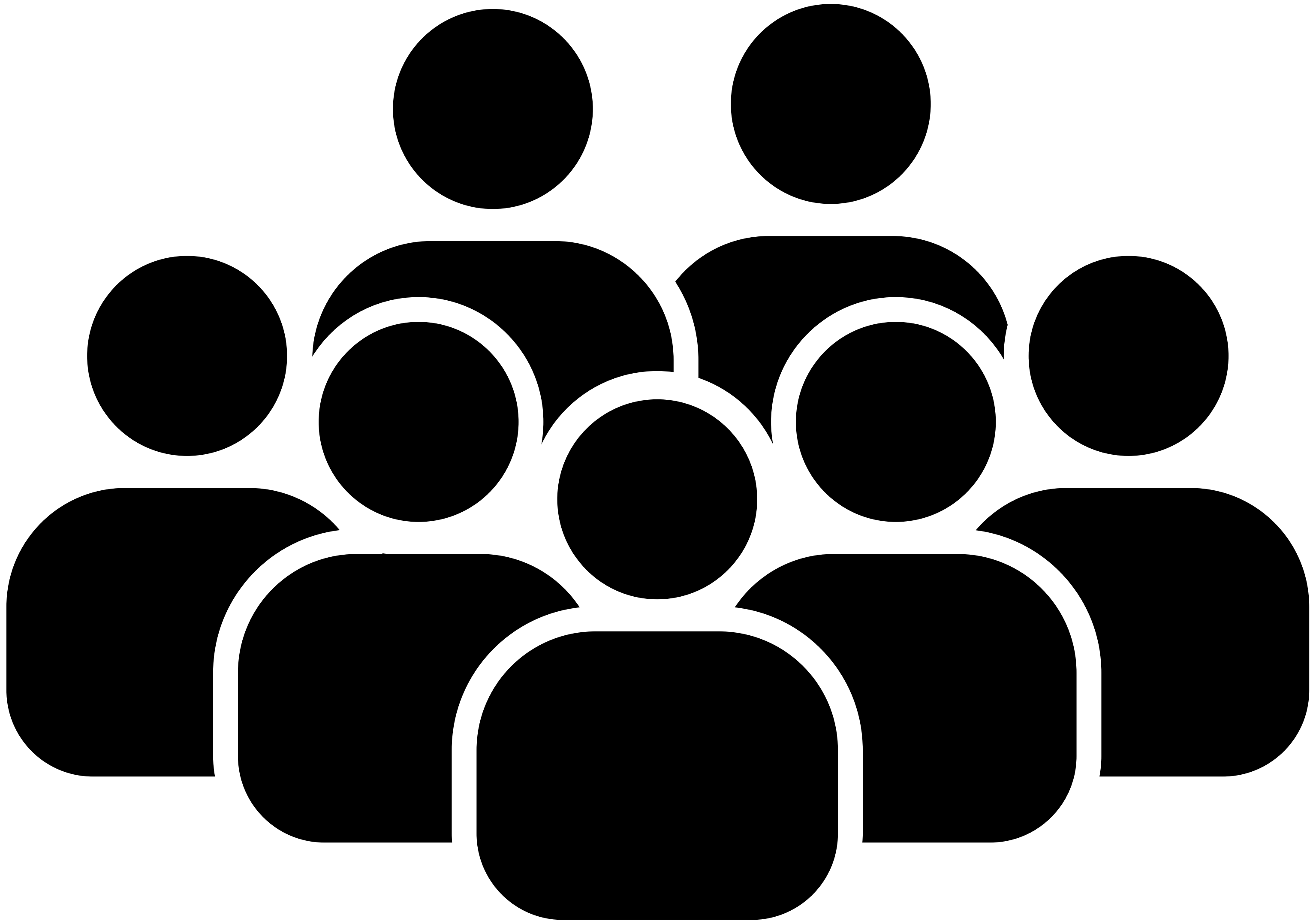
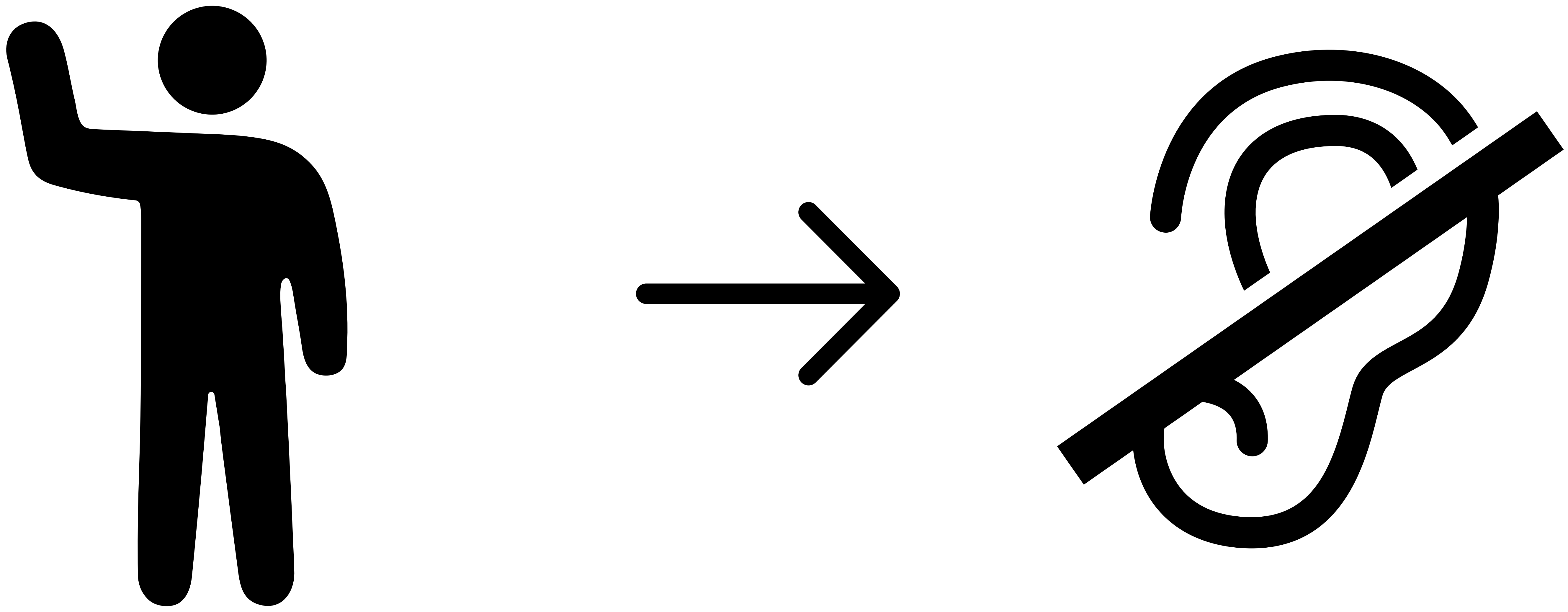
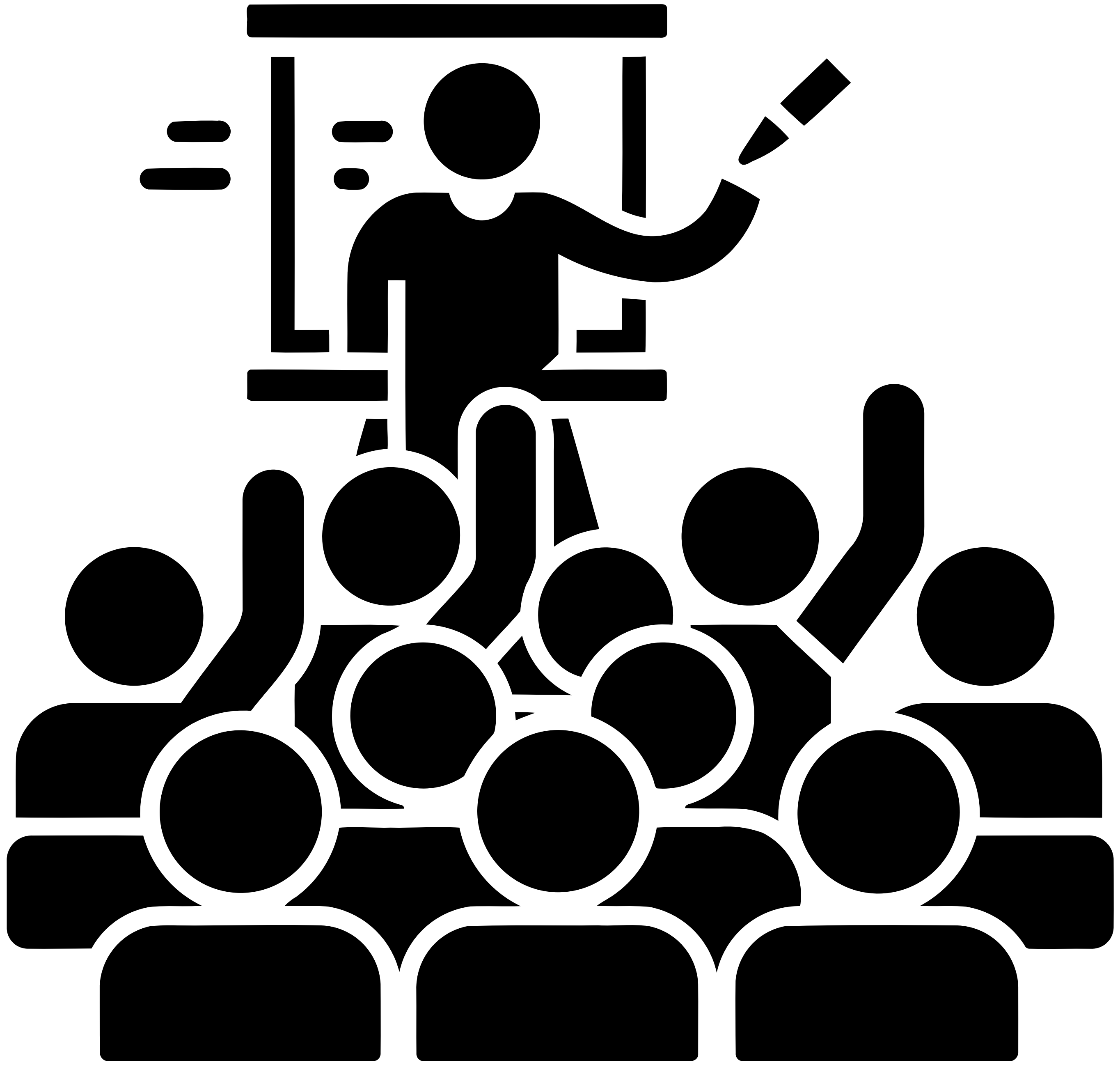
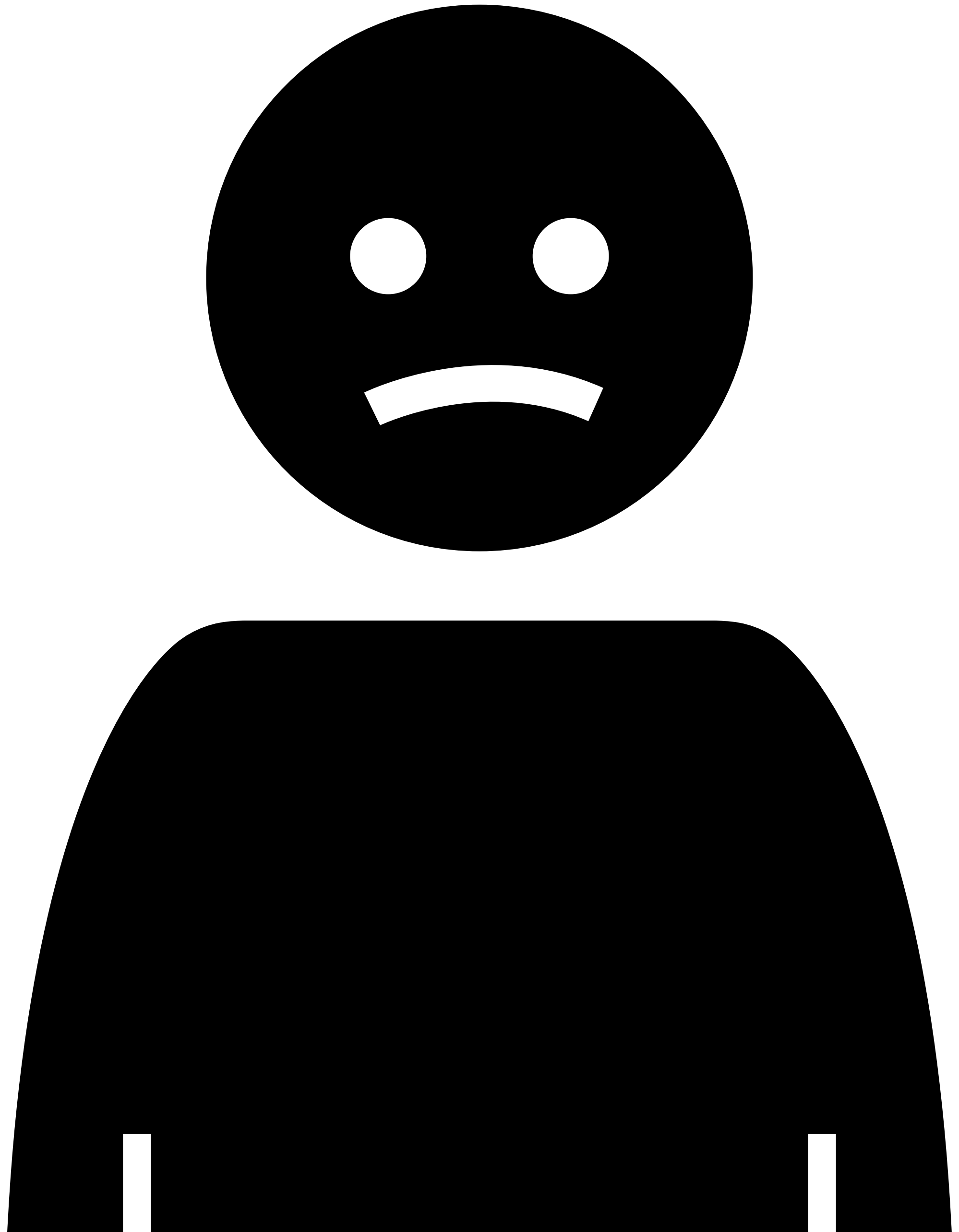
IBM

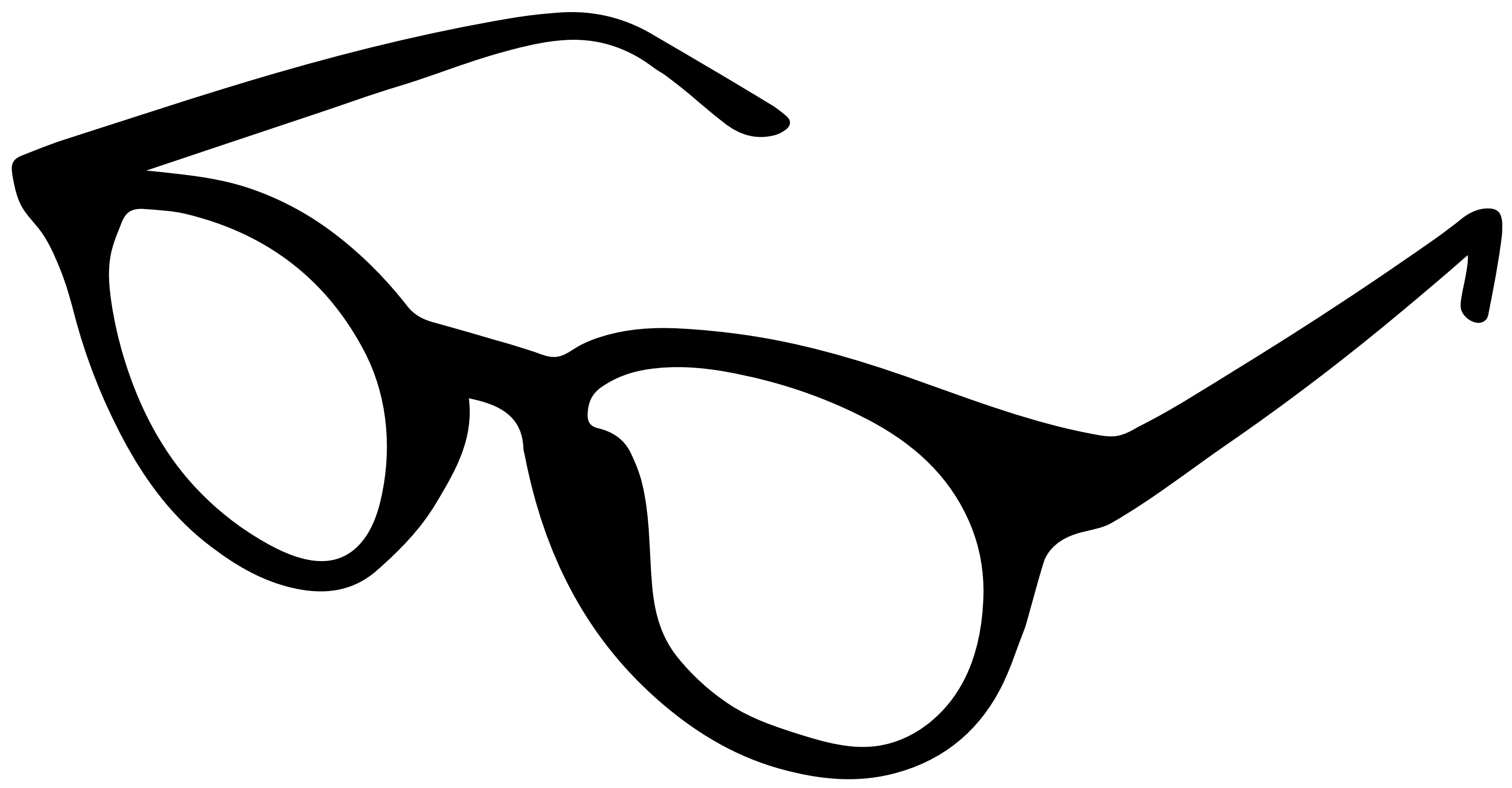
TLens: Speech Projection Glasses to Assist People with Hearing Impairments.

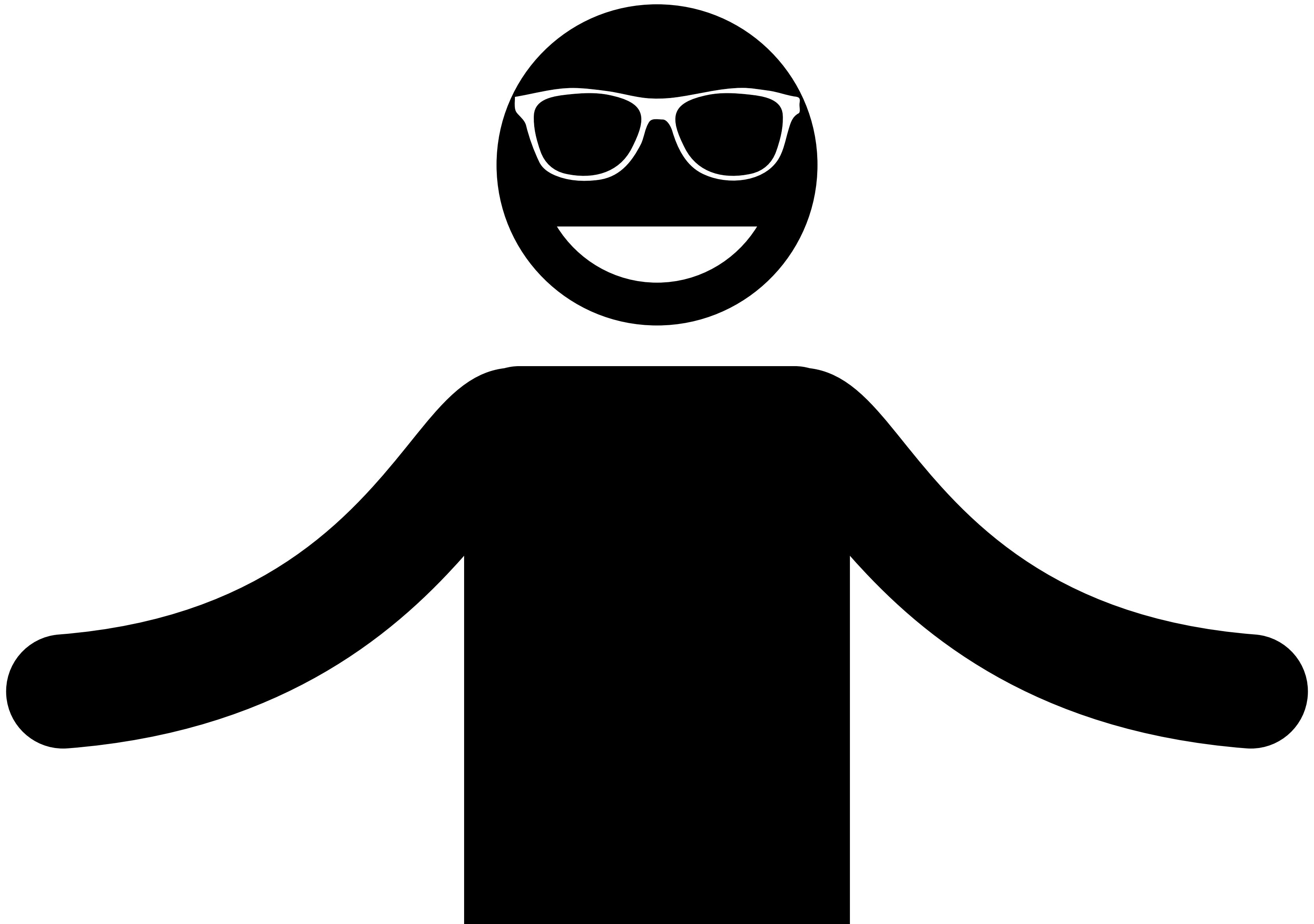












Our Team

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TLens

TLens

What Is TLens?

An Internet of Things (IoT)-based device that uses low-cost speech-to-text projection glasses to assist people with hearing impairments.

Specific Objectives

Optical System

Provide real-time subtitle projection directly on the lenses, ensuring clarity and easy viewing.

Speech Capture and Conversion

Convert spoken language into text accurately and instantly using a dedicated microphone and AI processing.

Management application

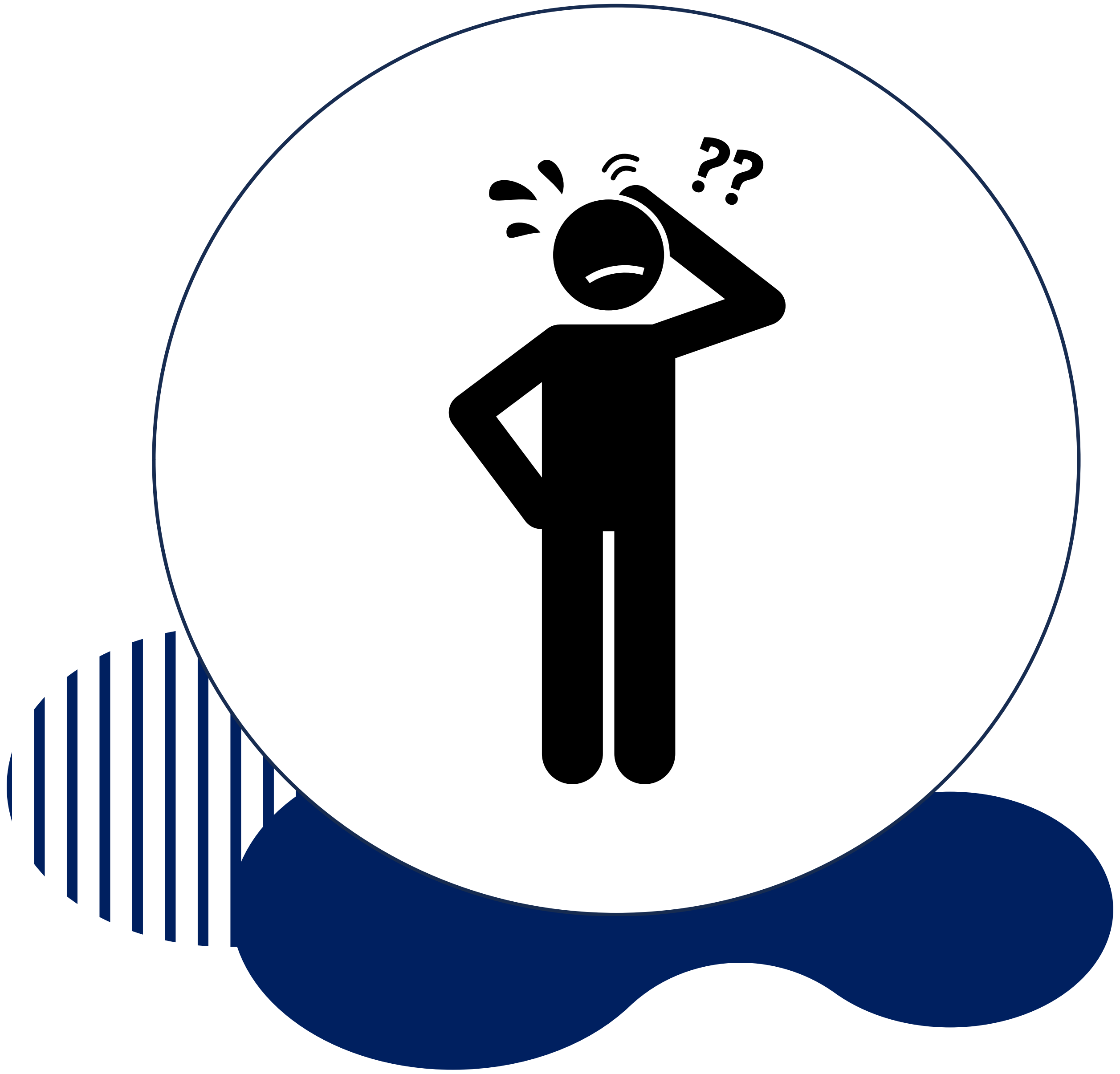
Allow users to configure, personalize, and operate the device through an intuitive mobile app.

Justification

TLens was created to reduce communication barriers for people with hearing loss. Current speech transcription technologies are expensive and not very accessible, so our goal is to offer a practical, low-cost, and inclusive solution.

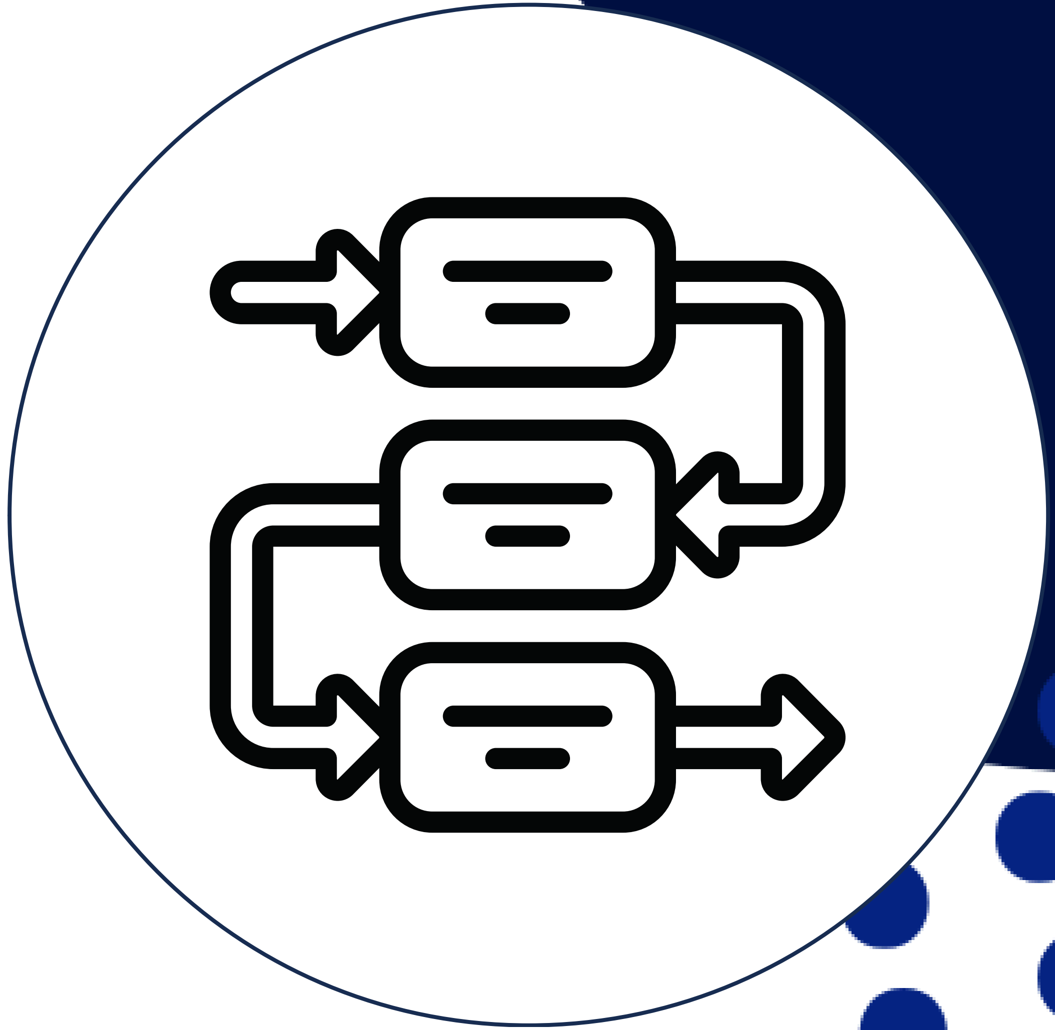
Main Challenge

The biggest challenge is social exclusion, since most people do not know sign language and communication often depends on others.

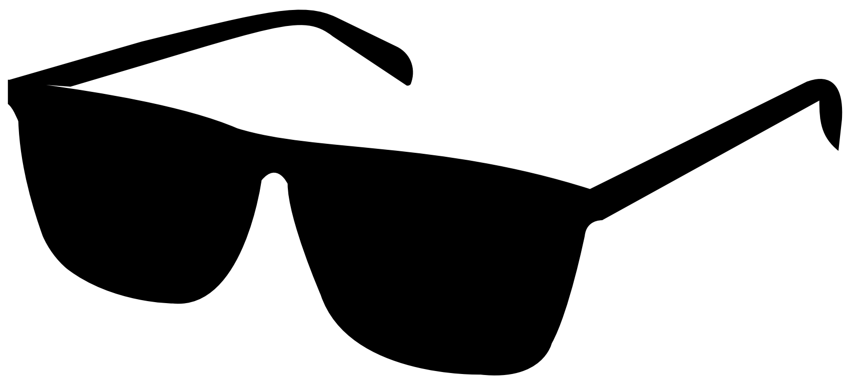


Methodology

To achieve our objective, we used methodologies with a qualitative approach, focused on perceptions and attitudes, according to Lakatos & Marconi (2003).

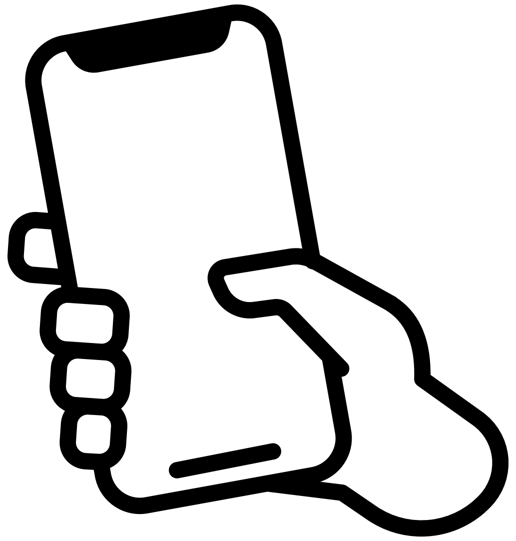


Main Concepts



IoT (Internet of Things)

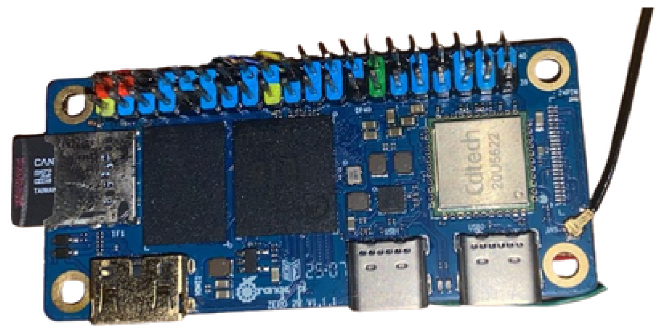
Integration of smart devices to make the solution more dynamic and connected, expanding the possibilities of interaction and accessibility.



Accessible Design

Focused on user experience, ensuring that deaf users have a clear, simple, and inclusive interface that facilitates communication in different contexts.

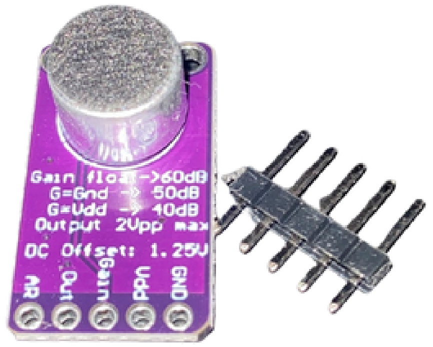
Technologies



Source: Own authorship, 2025.

Orange Pi Zero 2W

A compact development board with Wi-Fi and Bluetooth that processes audio and runs the system software.



Source: Own authorship, 2025.

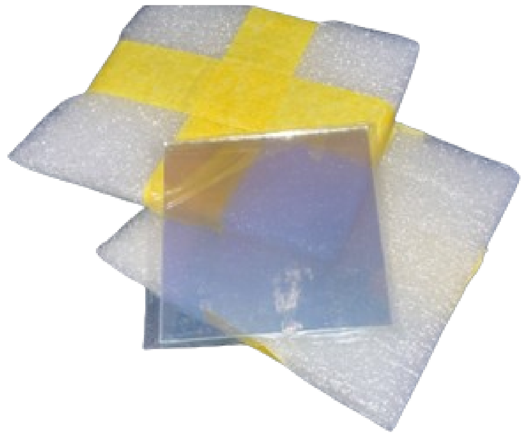
Microphone AGC

High-sensitivity audio module with automatic gain control to capture clear speech without distortion

Source: Own authorship, 2025.

OLED Display

A low-power I2C screen responsible for receiving and visualizing the transcribed text for projection.



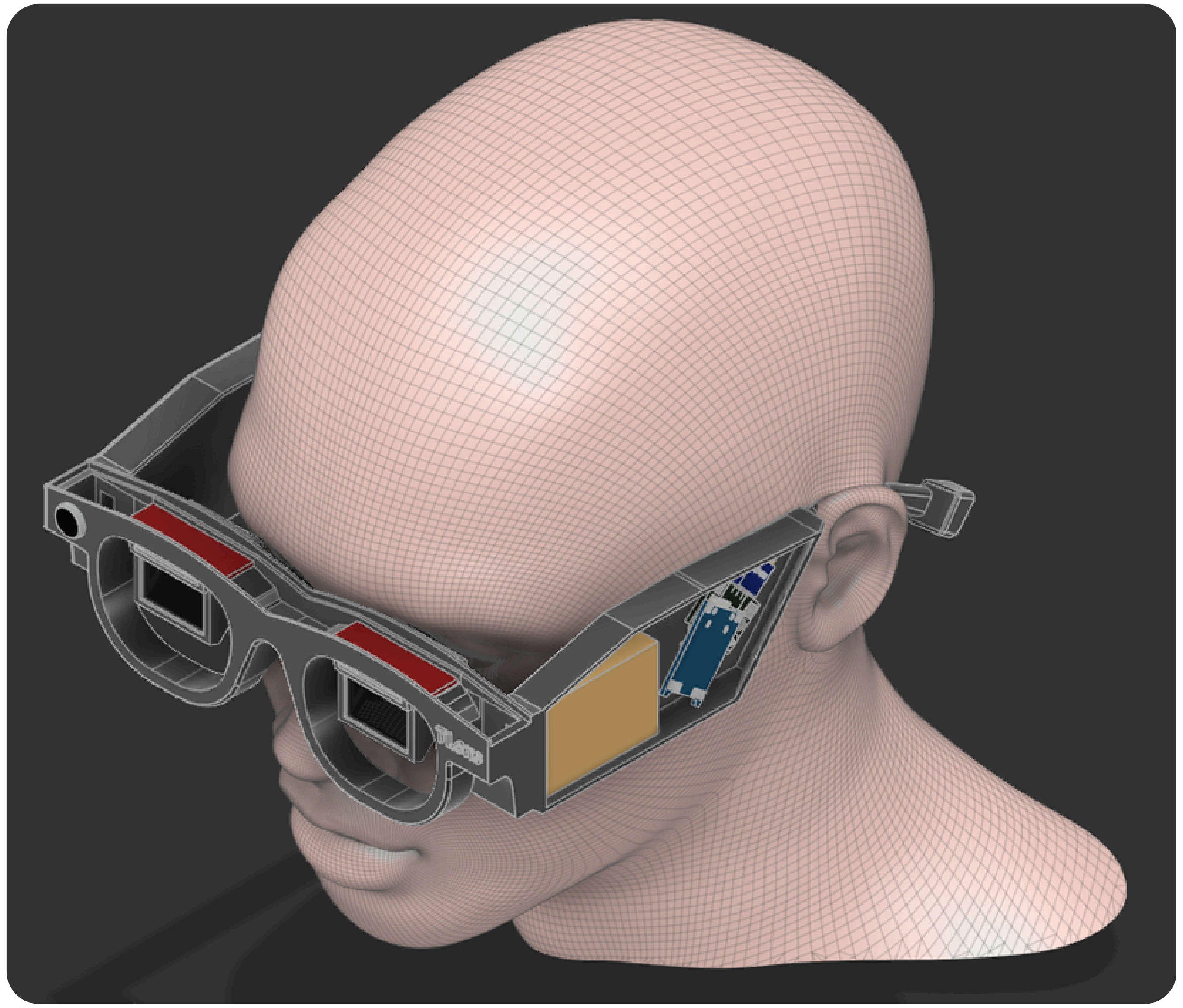
Source: Own authorship, 2025.

Semi-reflective lenses

Beam-splitting glass using Bird Bath optics to overlay text onto the user's view while maintaining transparency.

TLens

Expected Prototype



Source: Own authorship, 2025.

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TLens


TLens
Escolha seu idioma
Choose your language
Selecione o idioma que deseja usar no TLens.
Select your preferred language to use TLens
English
Português
Continuar

Considerations

We believe that technology should be a tool to help people. Our project connects theory and practice to promote inclusion. As we say: **Communication is a right, not a privilege.**

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References

- IBGE. **Estatísticas sobre deficiência no Brasil:** Pesquisa Nacional de Saúde 2019. Rio de Janeiro: Instituto Brasileiro de Geografia e Estatística, 2021. Disponível em: https://www.ibge.gov.br. Acesso em: 27 maio 2025.
- GELEDÉS. **IBGE confirma:** surdez não é sinônimo de Libras. Disponível em: https://www.geledes.org.br/ibge-confirma-surdez-nao-e-sinonimo-de-libras/. Acesso em: 27 maio 2025.Geledés+1LinkedIn+1
- HAND TALK**. Línguas mais faladas no Brasil:** será a Libras uma delas? Disponível em: https://www.handtalk.me/br/blog/linguas-mais-faladas-no-brasil/. Acesso em: 27 maio 2025.
- SANTOS, Daniel Lucas de Oliveira et al. **Desafios enfrentados para o atendimento de pacientes surdos: um estudo com docentes de Medicina.** Revista da Sociedade Brasileira de Fonoaudiologia, v. 49, n. 1, e027, 2025. Disponível em: https://doi.org/10.1590/1981-5271v49.1-2024-0175.
- OLIVEIRA, Thais S. de; et al. **A inclusão de deficientes auditivos no sistema público de saúde.** Revista Foco, v. 11, n. 1, p. 57-66, 2024. Disponível em: https://ojs.focopublicacoes.com.br/foco/article/view/6791. Acesso em: 18 maio 2025.
- TORI, Romero; HOUNSELL, Marcelo da Silva (org.). **Introdução a Realidade Virtual e Aumentada.** 3. ed. Porto Alegre: Editora SBC, 2020.
- ALIEXPRESS. Produto Lentes Semi Refletivas Divisoras de Feixe. Disponível em: https://pt.aliexpress.com/i/4001321844982.html Acesso em: 27 ago. 2025
- (Lakatos & Marconi, 2003)

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